

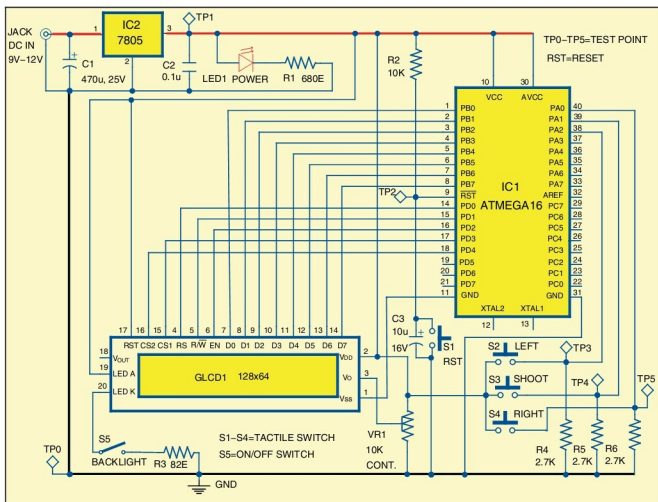


Fig. 3: Circuit of Space Invaders hand video game

Programmers Notepad. The program creates animated objects such as insects (invaders), a tank or spaceship, and an animated laser weapon that moves up to kill the invaders.

The code is basically an arrangement of arrays (consisting of byte codes of images generated). A byte code is integer (0-255) byte of data whose binary value shows the on/off status of each pixel across each page (8 pixels in a column). It is this arrangement of byte code that generates images on the LCD screen by activating or deactivating the pixels. Different arrays containing byte codes for different images are stored in the program memory (16 kB for ATmega16). Normal declaration of arrays is stored in the RAM (1 kB for ATmega16).

The next part is different function modules that contain codes for gener-

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	5V when key pressed
TP3	5V when key pressed
TP4	5V when key pressed
TP5	5V when key pressed

ating the byte codes from arrays on the screen. For example, tank() function is responsible for displaying the byte code for tank[] array, while insect1() function is used for ins[] array. They display graphics on the screen from the byte codes.

The next part is the use of interrupts in the game. Timer0 overflow interrupt used in the program serves two purposes: First, the interrupt being global, it isolates itself from the main

PARTS LIST

Semiconductors:

- IC1 - ATmega 16 microcontroller
- IC2 - 7805, 5V regulator
- LED1 - 5mm LED
- GLCD1 - 20-pin 128x64 GLCD with KS0108 controller

Resistors (all 1/4-watt):

- R1 - 680-ohm
- R2 - 10-kilo-ohm
- R3 - 82-ohm
- R4-R6 - 2.7-kilo-ohm
- VR1 - 10-kilo-ohm

Capacitors:

- C1 - 470µF, 25V electrolytic
- C2 - 0.1µF ceramic
- C3 - 10µF, 16V electrolytic

Miscellaneous:

- S1-S4 - Push-to-on switch
- S5 - On/off toggle switch
- CON1 - DC jack connector

loop time delays and generates interrupts at fixed intervals, serving as a master clock. Second, it implements the tasks that are not present in the main loop. The use of interrupts has been limited to only one because interrupts